

```
In [1]: import pandas as pd
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```
In [2]: dataset=pd.read_csv('insurance_pre.csv')
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```
In [3]: dataset
```

```
Out[3]:
```

	age	sex	bmi	children	smoker	charges
0	19	female	27.900	0	yes	16884.92400
1	18	male	33.770	1	no	1725.55230
2	28	male	33.000	3	no	4449.46200
3	33	male	22.705	0	no	21984.47061
4	32	male	28.880	0	no	3866.85520
...	...	...	...	...	...	...
1333	50	male	30.970	3	no	10600.54830
1334	18	female	31.920	0	no	2205.98080
1335	18	female	36.850	0	no	1629.83350
1336	21	female	25.800	0	no	2007.94500
1337	61	female	29.070	0	yes	29141.36030

1338 rows × 6 columns

```
In [4]: dataset=pd.get_dummies(dataset,drop_first=True)
```

```
In [5]: dataset
```

```
Out[5]:
```

	age	bmi	children	charges	sex_male	smoker_yes
0	19	27.900	0	16884.92400	0	1
1	18	33.770	1	1725.55230	1	0
2	28	33.000	3	4449.46200	1	0
3	33	22.705	0	21984.47061	1	0
4	32	28.880	0	3866.85520	1	0
...	...	...	...	...	...	...
1333	50	30.970	3	10600.54830	1	0
1334	18	31.920	0	2205.98080	0	0
1335	18	36.850	0	1629.83350	0	0
1336	21	25.800	0	2007.94500	0	0
1337	61	29.070	0	29141.36030	0	1

1338 rows × 6 columns

```
In [6]: dataset.columns
```

```
Out[6]: Index(['age', 'bmi', 'children', 'charges', 'sex_male', 'smoker_yes'], dtype='object')
```

```
In [8]: independent=dataset[['age', 'bmi', 'children', 'sex_male', 'smoker_yes']]
```

```
In [9]: independent
```

```
Out[9]:
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	age	bmi	children	sex_male	smoker_yes
0	19	27.900	0	0	1
1	18	33.770	1	1	0
2	28	33.000	3	1	0
3	33	22.705	0	1	0
4	32	28.880	0	1	0
...	...	...	...	...	...
1333	50	30.970	3	1	0
1334	18	31.920	0	0	0
1335	18	36.850	0	0	0
1336	21	25.800	0	0	0
1337	61	29.070	0	0	1

1338 rows × 5 columns

```
In [10]: dependent=dataset[['charges']]
```

```
In [11]: from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(independent,dependent,test_s
size=0.30,random_state=0)
```

```
In [23]: from sklearn.ensemble import RandomForestRegressor
regressor=RandomForestRegressor(n_estimators=1000,random_state=0)
regressor.fit(x_train,y_train)
```

C:\Users\GOD\Anaconda3\lib\site-packages\ipykernel_launcher.py:3: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

This is separate from the ipykernel package so we can avoid doing imports until

```
Out[23]: RandomForestRegressor(bootstrap=True, criterion='mse', max_depth=None,
                                max_features='auto', max_leaf_nodes=None,
                                min_impurity_decrease=0.0, min_impurity_split=None,
                                min_samples_leaf=1, min_samples_split=2,
                                min_weight_fraction_leaf=0.0, n_estimators=1000,
                                n_jobs=None, oob_score=False, random_state=0, verbose
                                =0,
                                warm_start=False)
```

```
In [24]: y_pred=regressor.predict(x_test)
```

```
In [25]: from sklearn.metrics import r2_score
r_score=r2_score(y_test,y_pred)
```

```
In [26]: r_score
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```
Out[26]: 0.8542190980526684
```

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In [ ]:
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In [ ]:
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